

Midterm quiz – Geospatial Data Science

For each question, exactly one answer is correct.

The general definition of a neighbor of an object i is:

- 1) an object next to i
- 2) an object closer to i than a threshold
- 3) an object with non-zero weight in column i of the weight matrix
- 4) an object with non-zero weight in row i of the weight matrix

Given a standardized spatial weight matrix, the spatial lag of an object i is:

- 1) the average of the values for all neighbors j of i
- 2) the sum of the values for all neighbors j of i
- 3) the sum of the reciprocal values for all neighbors j of i
- 4) the sum of all values minus the average distance

Assume a standardized spatial weight matrix with rook contiguity.

20	10	20
10	60	12
20	8	20

Then the spatial lag of the central cell is:

- 1) 180
- 2) 120
- 3) 100
- 4) 40
- 5) 15
- 6) 10

Assume a non-standardized spatial weight matrix with queen contiguity.

20	10	20
10	60	12
20	8	20

Then the spatial lag of the central cell is:

- 1) 180
- 2) 120
- 3) 100
- 4) 40
- 5) 15
- 6) 10

A spatial weight matrix W cannot be row-standardized when:

- 1) there is a row in W with only zeros
- 2) not all rows in W sum up to 1
- 3) there are disconnected clusters
- 4) W is asymmetric

When Moran's I is positive, it means:

- 1) there is a significant number of clusters with similar values
- 2) there is a significant number of clusters with dissimilar values
- 3) object and neighbor values are more similar than randomly reshuffled values
- 4) object and neighbor values are less similar than randomly reshuffled values

A low p-value for Moran's I means:

- 1) the observed I is likely to be measured in random reshuffling
- 2) the observed I is unlikely to be measured in random reshuffling
- 3) clusters are small
- 4) there is strong clustering